William Chuang

Tucson, AZ

williamhschuang@gmail.com   williamchuang.github.io

Summary of Qualifications

- **Strategic Thinker (INTJ)**: Focused on long-term, high-impact solutions in AI-driven cryptanalysis, OSINT, and cybersecurity.
- **Advanced Math & Physics:** Deep expertise in hyperbolic geometry, Kleinian groups, Galois theory, crucial for secure algorithm design.
- **Robust R&D:** 10+ years at respected institutions (Arizona, SFSU, Penn State, NTU), applying theoretical methods to practical challenges.
- **Security & Cryptography:** Applied curvature, limit sets, and Galois groups in post-quantum encryption and secure data protocols.
- **OSINT & Cybersecurity Research:** Self-studying Open-Source Intelligence (OSINT), pentesting, network forensics, and adversarial ML applications in threat detection.
- **Machine Learning & Data Security:** Applying ML models to system log analysis, TCP/IP traffic inspection, firmware integrity verification, and network anomaly detection.
- **Computational Proficiency:** Python (NumPy, scikit-learn, PyTorch), R, C/C++, Java, Lisp, Mathematica, Linux/Bash scripting, and reverse-engineering low-level firmware; built large-scale cryptography tools.

Education

University of Arizona (Expected Spring 2025)
M.S. in Mathematics, advanced Ph.D.-level coursework

San Francisco State University (Spring 2022)
M.A. in Mathematics; Thesis on Schottky groups (Advisor: Dr. C.-K. Lai)

University of San Francisco (Fall 2018)
B.S. in Mathematics, Minor in Computer Science, GPA: 3.88/4.00, Honors

Core Competencies

Mathematical Cryptography & Security:
Hyperbolic geometry, Kleinian groups, post-quantum cryptography, topological vulnerabilities.

Developing a novel encryption framework leveraging group actions on neural networks rather than solutions of differential equations. This approach, independently conceived and later found to parallel the motivation behind Poincaré's Fuchsian groups and monodromy, restructures security landscapes in the post-quantum era by embedding cryptographic transformations within geometric symmetries.

Data Analysis & Machine Learning:
Transformer architectures (multi-head attention), autoencoders, geometric/topological ML approaches.

Algorithm & HPC Development:
Python/C++ for large-scale data, GPU-accelerated ML, system-level optimization.

Research & Technical Writing:
Multiple publications/presentations; formal proof tools (Lean 4, LLMs).

AI-Augmented Intelligence Analysis & OSINT Automation:
Developing transformer-based (local LLM) AI agents trained on intelligence cycle methodologies (e.g., Heuer's structured analysis, HUMINT techniques) for real-time data fusion, deception detection, and adversarial influence modeling.

Human Factor Security & OSINT:
Defending against psychological manipulation, decoding body language, behavioral profiling, lie detection via baseline analysis, NLP for communication security, countering mind control tactics, and influence operations in intelligence settings.

Cognitive & Influence Security:
Adversarial intelligence analysis, AI-enhanced OSINT gathering, NLP-driven deception detection, psychological profiling via transformer models, and social engineering countermeasures.

Relevant Research Experience

University of Arizona (2022–Present)

- *Prof. S. Sethuraman:* Real analysis; self-studied stochastic processes for cryptanalysis.

- *Prof. S. Cherkis:* Explored Nahm equations, geometric field theories, and used Lean 4+LLMs for secure AI.
- *Prof. N. Hao:* RTG Project on transformer attention scaling.
- *Prof. C. Haessig:* Investigated corresponding polynomials of Galois gorups by writing Python code, self-studying this for cryptographic classification.
- *Prof. D. Glickenstein:* Mentored a project reconstructing Mirzakhani's study on hyperbolic geometry and closed geodesics, with self-study on its application for encryption using transformer architectures and autoencoders.

San Francisco State University (2019–2022)

- Computed Hausdorff dimension of Schottky groups; applied fractal geometry for data obfuscation.
- Applied the prime geodesic theorem to secure high-dimensional data.

Pennsylvania State University (2017–2018)

- Investigated Hardy's proof of uniform distribution (pseudo-random generation).
- Studied topological invariants for encryption algorithms.

NTU—LeCosPA (Pre-Baccalaureate, 2011–2013)

- Researched TQFT, AdS/CFT, and vacuum energy; early work in quantum information.

Additional Research Projects

Self-Study: Semisimple Rings and Radicals in Coding Theory and Cryptography
Based on Prof. Klaus M Lux's lecture (Spring 2025), this project explores the role of semisimple rings and the Jacobson radical in coding theory and cryptography. Topics include linear codes over rings (e.g., $\mathbb{Z}_4$), group algebras, ring-based cryptosystems (NTRU, Ring-LWE), and ten concrete examples illustrating how nontrivial radicals influence cryptographic security.

Self-Study: AI-Driven Network & Firmware Security Analysis
Applying machine learning to detect anomalies in system logs, TCP/IP traffic, and firmware integrity. Researching adversarial ML techniques for cyber threat detection, focusing on vulnerabilities in processors (e.g., AMT), chipsets, and network traffic patterns. Investigating anomaly detection across all OSI layers and forensic-level system monitoring.

Self-Study: RF Signal Analysis for SIGINT & Cybersecurity
Applying spectrum analysis, software-defined radio (SDR), and machine learning to detect, classify, and mitigate anomalous RF signals, including the detection of unauthorized cellular simulators (such as IMSI catchers, rogue base stations, and spoofed cell towers used by malicious actors for surveillance, data injection, or signal jamming). Researching RF propagation, modulation techniques, and counter-SIGINT strategies.

Self-Study: AI-Driven Cognitive Security & OSINT Defense
Designing AI-powered analyst teams using transformers (local LLMs) for autonomous intelligence gathering, behavioral profiling, and deception detection. Researching automated influence analysis and adversarial social engineering defense using Heuer's methods (Analysis of Competing Hypotheses, Structured Analytical Techniques).

Self-Study: AI-Driven Narrative Simulation & Influence Modeling
Researching AI-driven frameworks for strategic narrative generation, adversarial influence simulations, and predictive modeling of geopolitical discourse. Developing transformer-based AI for policy impact analysis, disinformation detection, and automated strategic communications. Exploring AI-enhanced wargaming environments to test leadership messaging strategies, adversarial competition, and cognitive warfare principles. Studying structured analytic techniques (SATs), psychological operations (PsyOps), and counter-deception models to advance information dominance in government, defense, and corporate intelligence applications.

Emerging Sensing Technologies & Counter-Surveillance:
Self-studied laser-based Non-Line-of-Sight (NLoS) imaging capable of reconstructing rooms through keyholes or occlusions. Currently researching counter-surveillance techniques using hyperbolic transformers to detect or interfere with adversarial NLoS imaging. Exploring future applications to human-computer interaction (HCI) via remote gesture sensing through walls and barriers. Emphasizing dual-use applications for both intelligence collection (next-generation surveillance) and defensive countermeasures (anti-surveillance and physical security augmentation).

Self-Study: Multimodal Environmental Sensing — RF, Ultrasound, & IoT Fusion for Privacy, Healthcare, HCI, & Real-Time Polymath Workflows
Researching multimodal environmental sensing combining WiFi-based human pose estimation (RF DensePose), supersonic/ultrasound sensing using smartphone and IoT device speakers/microphones, and cooperative swarm sensing across distributed devices. Applying hyperbolic transformers for both sensing enhancement and privacy-preserving countermeasures.
This study spans:

- **Privacy-Preserving Counter-Surveillance**: Developing adversarial signal injection and hyperbolic signal shaping techniques to defend against unauthorized through-wall or through-device imaging by adversarial actors.
- **Home Healthcare & Elderly Monitoring**: Real-time non-contact monitoring using WiFi signals, ultrasound echoes from smartphones/IoT speakers, and fusion across environmental sensors to detect

posture, falls, and respiratory patterns.

- **First Responder & Disaster Recovery**: Using RF + ultrasound hybrid sensing for detecting human presence through smoke, debris, or walls in collapsed buildings or emergency rescue operations.
- **Global Anatomical Data for AI Surgical Training**: Passive collection of anonymized human motion and anatomical movement data across populations using everyday devices, building a global real-time anatomical dataset to train AI surgeons and autonomous medical robots.
- **Human-Computer Interaction (HCI) & Real-Time Polymath Workflows**: Developing gesture/movement-based control interfaces using combined RF, ultrasound, and environmental sensing — enabling seamless, touchless interaction across AR/VR workspaces, smart homes, autonomous vehicles, and advanced research environments where rapid switching between scientific, mathematical, intelligence, and technical disciplines is required.

This integrates:

- **Wireless Physical Layer Security** (RF/ultrasound channel obfuscation)
- **Hyperbolic Geometry for Signal Processing**
- **AI-Driven Multimodal Sensor Fusion**
- **Medical Imaging via Passive Sensing**
- **Edge AI on Smartphones, IoT, and Wearables**
- **Cognitive Ergonomics for Real-Time Polymath HCI**

Applications span:

- Privacy-preserving smart home environments (smartphones doubling as silent guardians).
- Telemedicine with non-contact remote diagnostics.
- First responder deployment in smart disaster zones.
- HCI innovations for polymath researchers needing seamless transitions between symbolic math, technical coding, OSINT research, and physical prototyping.
- Spatial computing interfaces for intelligence analysis, multi-modal sensemaking, and collaborative situational awareness.

Self-Study: Transformer-Based Analysis & Countermeasures for EM/Stealth Radiation Document Espionage

Investigating advanced espionage techniques where adversarial actors deploy combined radiation and electromagnetic (EM) waves—ranging from terahertz and zeta waves to backscatter and passive reflectometry—to covertly scan printed documents on desks, bookshelves, or in file cabinets, retrieving content without physical access.

This research integrates:

- Applying transformer-based AI to reconstruct potential document exposures by reverse-engineering scanned data footprints.
- Developing hyperbolic transformer architectures to detect signal anomalies, leveraging curvature-sensitive embeddings to highlight unnatural EM reflections characteristic of covert scanning.
- Applying SDR (Software-Defined Radio) and real-time spectrum analyzers to continuously monitor the local EM environment, detecting covert scanning transmissions in real time.
- Exploring adversarial radiative shielding using AI-optimized metamaterials, structured electromagnetic interference (EMI), and dynamic scattering techniques to actively distort or mask document content.
- Formulating inverse problems to infer adversarial scanner configurations from detected emission patterns, enabling real-time counterintelligence alerts for classified or proprietary document protection.
- Integrating findings into comprehensive counter-surveillance and physical security intelligence (PHYSINT) toolkits, linking document-level monitoring with broader facility security intelligence.

Potential Applications:

- Protection of classified documents, trade secrets, and sensitive archives in government, defense, and corporate environments.
- Continuous EM environment monitoring for SCIFs, executive boardrooms, and high-risk offices.
- Design of next-generation secure office spaces, blending AI-guided shielding, active jamming, and smart concealment within physical spaces.

This project advances Prodeo's future capabilities in AI-enhanced counterintelligence, blending SIGINT, PHYSINT, and adversarial ML into adaptive counter-surveillance systems.

Self-Study: Advanced Defense Systems — Next-Generation Projectiles & Subsurface Security

Researching emerging methods for precision-guided projectiles, terrain-penetrating munitions, and hybrid-domain deterrence systems. Focus includes:

- Physics and engineering of projectile-launch mechanisms leveraging electromagnetic, plasma-assisted, and hybrid propulsion technologies.
- Subsurface security challenges including adversarial tunneling, underground domain awareness, and sensor fusion.
- AI-enhanced modeling for optimizing projectile trajectories across air, water, and solid earth.
- Integration of under-terrain sensing, autonomous subterranean drones, and real-time countermeasures within broader C4ISR ecosystems.
- Cross-domain fusion of subsurface, underwater, and near-space kinetic capabilities for next-generation integrated deterrence.

Potential Applications:

- Strategic infrastructure protection (nuclear silos, undersea cables, communication hubs)
- Defense against adversarial tunneling, mining, and subsurface infiltration
- Rapid-response countermeasure systems for critical asset protection
- Next-generation coastal and inland defense blending subsurface, surface, and aerial ISR

Note: This self-study also supports future consulting capabilities under Prodeo, offering dual-use solutions for defense clients, critical infrastructure operators, and advanced engineering teams exploring subsurface and hybrid-domain challenges.

Self-Study: Counter-Detection for Terrain-Penetrating Munitions Using Gravitational Waves & Neutrino Imaging

Researching advanced sensor and detection systems leveraging gravitational waves and neutrino-based sensing to detect and map terrain-penetrating munitions and subsurface threats. This approach addresses current limitations in radar, seismic, and acoustic detection by:

- Exploring gravitational perturbations caused by projectile movement or underground excavation, detectable via space-based or ground-based gravitational wave sensors.
- Investigating neutrino backscattering or diffraction signatures from dense subterranean objects (e.g., munitions, tunnels, or reinforced structures).
- Applying hyperbolic transformer-based AI to fuse multi-physics sensor data into coherent, real-time subsurface threat models.
- Developing anomaly detection techniques for long-duration monitoring, highlighting subtle gravitational or neutrino signal deviations against natural background noise.
- Studying feasibility of distributed sensor networks (space-ground hybrid) for persistent monitoring of critical subsurface assets, infrastructure, and border regions.

Potential Applications:

- Early warning and tracking of deep-penetration munitions and adversarial excavation efforts.
- Dual-use scientific sensors for both astrophysical research and applied national security.
- Real-time subsurface "neutrino radar" for next-generation underground domain awareness.

Note: This study supports Prodeo's future capabilities in dual-use subsurface ISR and gravitational/neutrino-based intelligence applications, relevant to both defense clients and scientific partners.

Prodeo: AI-Driven Intelligence Solutions (Launching July 2025)

Founder & Independent Researcher

- Full operational launch scheduled for July 2025 to deliver specialized AI-driven intelligence solutions across OSINT, SIGINT, cybersecurity, cognitive security, and strategic influence modeling.
- Researching transformer-based AI for real-time cyber threat intelligence, geopolitical risk assessment, adversarial influence simulations, and disinformation detection.
- Developing autonomous AI teams for narrative analysis, predictive intelligence modeling, and multi-domain influence simulations, supporting applications for government, defense, and private sector intelligence needs.
- Designing AI-powered SIGINT analysis platforms, integrating spectrum monitoring, SDR-based signal intelligence, and anomaly detection in RF environments to monitor adversarial communication and detect covert transmissions.
- Engineering AI-enhanced forensic cybersecurity systems for end-to-end monitoring — including system logs, TCP/IP network traffic, firmware integrity checks, and anomaly detection across all OSI layers.
- Researching hyperbolic transformer-based techniques to protect sensitive physical documents from adversarial EM/radiation-based document espionage (terahertz/zeta wave scanning, backscatter imaging). Developing AI-assisted shielding using metamaterials and dynamic electromagnetic interference (EMI) cloaking.
- Combining AI-augmented OSINT automation, psychological operations (PsyOps) modeling, and cognitive warfare research into real-time narrative analysis engines to predict and counter adversarial influence operations.
- Innovating in multimodal environmental sensing, integrating WiFi human pose estimation, ultrasound radar from mobile devices/IoT speakers, and RF/EM fusion for privacy-preserving surveillance countermeasures, real-time healthcare monitoring, and HCI innovations for polymath workflows.
- Researching laser-based Non-Line-of-Sight (NLoS) imaging (e.g., reconstructing rooms through keyholes) and its countermeasures, including hyperbolic-transformer-based anomaly detection and smart architectural shielding for SCIFs and executive spaces.
- Exploring plasma and EM-based propulsion and defensive systems for underwater, subsurface, and aerospace security — applying this to quiet submarine propulsion, subsurface defense against tunneling threats, and counter-space applications.
- Investigating next-generation projectiles (terrain-penetrating munitions) and developing gravitational wave and neutrino-based sensing techniques to detect and map subsurface threats — advancing dual-use capabilities for both strategic deterrence and scientific sensing.
- Applying structured analytic techniques (SATs), cognitive warfare methodologies, and AI-enhanced deception detection across all intelligence domains — integrating these into a unified AI decision-support platform for intelligence-driven operations.
- Actively developing proprietary LLM-based AI agents trained directly on the intelligence cycle (Heuer's struc-

tured analysis, HUMINT/SIGINT/OSINT fusion), enabling rapid sensemaking, red teaming, and adversarial simulations in both government and private sector environments.
- Preparing Prodeo to offer consulting, technology development, and AI-enhanced analytical solutions for national security clients, defense contractors, technology startups, and private sector organizations requiring intelligence automation, adversarial modeling, and strategic influence analysis.

Teaching & Leadership

University of Arizona (2022–Present): GTA for College Algebra/Calculus.

San Francisco State University (2019–2022): GTA for Calculus, focusing on proof-based exploration.

Awards & Certifications

- Protecting God's Children Online Awareness 4.0 – Order of Malta (March 2025)
- Nominated for MSRI Summer School, Oxford (Metric Geometry, 2021)
- Information Security Awareness & Safety Training, Univ. of Arizona (2023)
- MASS Scholarship, Penn State (Full Tuition, 2017)
- ACM SIGMOD Service Award (2016)
- Big Data Training, MIT CSAIL (2015)

Technical Skills

Programming & Tools: Python, C/C++, Java, Lisp, R, Mathematica, Shell, Lean 4, Git/GitHub, LaTeX.

Cybersecurity & OSINT:
AI-assisted network forensics, AI-driven anomaly detection in TCP/IP traffic, firmware-level intrusion detection, reverse-engineering network threats, cryptanalysis, penetration testing via transformer models.

System & Network Security: TCPDump/Wireshark analysis, intrusion detection, monitoring firmware/memory vulnerabilities, reverse-engineering Intel AMT and similar architectures for anomaly detection.

Mathematical & Computational Methods: Real/complex analysis, measure theory, topology, functional analysis, stochastic processes, encryption/decryption, HPC, advanced cryptography.

Radio Frequency (RF) & SIGINT: Spectrum analysis, Software-Defined Radio (SDR), RF signal detection, FCC amateur radio exam preparation.

AI-Enhanced Intelligence & OSINT:
AI-driven intelligence automation (Heuer-based LLM agents), adversarial social engineering defense, NLP-driven deception detection, multi-agent data fusion for SIGINT/HUMINT integration.

Additional Information

Faith: Catholic (Confirmed 28 years, e-Knight of Columbus, awaiting CUF exemplification). Completed Protecting God's Children Online Awareness 4.0 (March 2025) under the Order of Malta for safeguarding minors and vulnerable individuals.

Languages: English (Fluent), Mandarin/Taiwanese (Native), Learning Latin, French, Spanish, Italian, Hebrew

Memberships: Pi Mu Epsilon Honor Society (University of San Francisco)

References: Available upon request